A linear/non-linear adjunction of sheaves over a site of causal interaction contexts

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Extended abstract — machine-checked in cubical Agda

Abstract

We construct, with a complete machine-checked proof in cubical Agda, a sheaf-level linear/non-linear adjunction in the sense of Benton over a small site of causal interaction contexts. The objects of the site are finite local views of a concurrent computation — a community of unityped actors, the events they take part in, and the causal order on those events. The topology is the one generated by the decomposition of each context into its single events. We show that a presheaf on this site is a sheaf exactly when it is determined by its values at single events, and that the inclusion of these sheaves into presheaves is reflective: for any coefficient category with finite products, the reflector (sheafification, in the precise sense made explicit below) sends a presheaf to the product of its event-local values. Given a Benton linear/non-linear adjunction $F \dashv G$ between a cartesian category and a $*$-autonomous one at the level of coefficient categories, we lift it to sheaves and obtain an adjunction between cartesian-valued and linear-valued sheaves, with the linear exponential "!" recovered as the induced comonad.

This note isolates one machine-checked theorem from a larger programme that models communities of communicating agents — and the typed data their interaction produces — in the language of sheaves and adjoint functors. The programme rests on a deliberate *type discipline*: the agents themselves are unityped, indistinguishable to the system, while all structure is carried by the typed data they produce — session types in the orchestration layer this theorem concerns, other type disciplines in other layers. Its governing commitment is perspectival: there is no absolute, view-from-nowhere description of what a community of agents is doing, so any global description must be assembled from local ones. The theorem below is the first place that commitment is made precise on the operational side. It concerns a site whose objects are local snapshots of an interaction and a topology whose only covers are the decompositions of a snapshot into its individual events. In that setting a *sheaf* is precisely a description determined by what each single event sees, and *sheafification* is the canonical, universal way to manufacture such a global-from-local description out of any raw assignment of meanings to snapshots. The system's "global point of view" is thus not imposed from outside; it is the coherent reconstruction of its event-local standpoints. The second half of the theorem shows that this reconstruction is compatible with the resource-sensitive (session-typed) structure the typed data themselves carry.

The site

The actor model of concurrency [1] describes computation as a community of agents — *actors* — that interact solely by exchanging messages and that the system does not distinguish from one another: they are *unityped*. To this we add causal structure in the sense of Lamport [2]: the events

of an interaction carry a partial order recording which event may have influenced which. Following the tradition of presheaf models of concurrency [3], where concurrent behaviour is studied through (pre)sheaves on a small category of observations, we organize these local views into a site.

Fix a set A of actors — a bare set, with nothing distinguishing its elements, which is the formal content of the actors being *unityped*. A *finite causal context* over A is a finite set of events equipped with a partial order — the causal order — together with an incidence relation recording which actors take part in which events; we require the context to be *supported*, meaning that every actor present takes part in at least one event. A morphism of contexts is a *causality-reflecting embedding*: an embedding of one context as a sub-view of another that preserves actors, events, and incidence, and that both preserves and reflects the causal order. Write Site_A for the resulting category and call it the *actor-causal site*. This is our name for the object; its ingredients — unityped actors, a causal order on events, presheaf semantics — are standard, and the causal order is the one familiar from event structures in Winskel's sense.

The topology is generated by a single kind of cover: the decomposition of a context into its individual events. For an event e of a context F, let $\mathrm{pt}_e(F)$ be the one-event subcontext at e, whose actors are exactly those that e touches. The generating covering family of F is

$$\{\, \mathrm{pt}_e(F) \longrightarrow F \,\}_{e \in E(F)}.$$

We make no other covers primitive: causal refinements and higher overlaps of events are deliberately left to richer topologies. We call a presheaf satisfying descent for these covers a *point sheaf*. We stress at once what is and is not proved: descent is established for this explicitly presented family of single-event covers; identifying it with the sheaf condition for a fully formalized Grothendieck coverage in the sense of SGA 4 [4] is left to later work (see Limitations).

Sheafification

Let $\mathcal{C}$ be a *coefficient category* with finite products — the category in which our presheaves take their values, and where those values are the typed data the system assigns to a context. Write $\mathsf{PSh}_A(\mathcal{C}) = [\mathsf{Site}_A^{\mathrm{op}}, \mathcal{C}]$ for the $\mathcal{C}$-valued presheaves and $\mathsf{Sh}_A(\mathcal{C}) \subseteq \mathsf{PSh}_A(\mathcal{C})$ for the full subcategory of point sheaves. A presheaf is a point sheaf precisely when it is determined by its values at single events; the formalization proves this as an equivalence between descent for the canonical cover and a product-locality condition.

The inclusion $\iota_{\mathcal{C}} : \mathsf{Sh}_A(\mathcal{C}) \hookrightarrow \mathsf{PSh}_A(\mathcal{C})$ has an explicit left adjoint. For a presheaf P, set

$$a_{\mathcal{C}}(P)(F) = \prod_{e \in E(F)} P\big(\mathrm{pt}_e(F)\big),$$

with restrictions induced by the point-to-point morphisms. This is the *sheafification* functor for the point topology — the reflector exhibiting point sheaves as a reflective subcategory of presheaves. We use the word "sheafification" in exactly this restricted sense, and no other: $a_{\mathcal{C}}$ is the left adjoint to the inclusion of the presheaves that satisfy descent for the canonical single-event covers. We do not claim it is the associated-sheaf functor for a fully formalized coverage. The construction is carried out behind an abstract finite-product interface, so that the proof assistant does not unfold the products into enumeration-dependent normal forms during unification. The machine-checked result is the adjunction

$$a_{\mathcal{C}} \dashv \iota_{\mathcal{C}}.$$

The linear/non-linear adjunction

The typed data produced by interaction are resource-sensitive: a message once consumed is not silently reused, a session once advanced does not revert. This is the domain of *linear logic*, and — because the word is badly overloaded — we mean linear *logic* throughout, never linear algebra. We take the linear coefficient category V to be a $*$-autonomous category [8], a model of multiplicative linear logic and the categorical home of session types in the sense of Caires and Pfenning [9]. The contrasting "non-linear" side is the cartesian fragment, in which typed data may be freely copied and discarded.

A *linear/non-linear adjunction* in the sense of Benton [7] consists of a cartesian category C, a symmetric monoidal (here $*$-autonomous) category V, and a monoidal adjunction $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$; the induced comonad $F \circ G$ on V is the linear exponential "!", the modality that re-admits the structural rules of copying and discarding. We assume such an adjunction at the level of coefficient categories, with both C and V carrying finite products.

Postcomposition lifts F and G to presheaves,

$$F_* : \mathsf{PSh}_A(\mathsf{C}) \to \mathsf{PSh}_A(\mathsf{V}), \qquad G_* : \mathsf{PSh}_A(\mathsf{V}) \to \mathsf{PSh}_A(\mathsf{C}),$$

with $F_* \dashv G_*$ computed pointwise. The right adjoint preserves point sheaves: descent along the canonical cover is transported across the natural bijection $\mathsf{C}(X, GY) \cong \mathsf{V}(FX, Y)$. The left adjoint is made to land in sheaves by composing with sheafification. Writing $R = G_*$ restricted to point sheaves and $L = a_{\mathsf{V}} \circ F_* \circ \iota_{\mathsf{C}}$, we obtain the following.

Theorem. *Let C be a cartesian category and V a $*$-autonomous category, both with finite products, and let $F \dashv G : \mathsf{C} \rightleftarrows \mathsf{V}$ be a linear/non-linear adjunction in the sense of Benton. Then there is an adjunction*

$$L \dashv R : \mathsf{Sh}_A(\mathsf{C}) \rightleftarrows \mathsf{Sh}_A(\mathsf{V})$$

between cartesian-valued and linear-valued point sheaves on the actor-causal site. The induced comonad $L \circ R$ is the sheaf-level linear exponential "!".

The result is formalized without postulates in the modules `SupportedChangeOfBase`, `SupportedLinearNonlinea` and `PointLinearNonlinearAdjunction`, building on the reflector constructed in `PointSheafAdjunction`.

Scope and significance

The theorem is deliberately modest, and three classical lineages meet in it. It is a descent statement over a small site, in the tradition of Grothendieck [4, 5]. It is an instance of Lawvere's principle that adjointness is the right way to organize foundational structure [6]: the global object is the value of a left adjoint, and its universal property is exactly the freeness of the reconstruction. And it lifts Benton's linear/non-linear adjunction from coefficients to sheaves. The conceptual content is that event-local observations *freely* generate a coherent global object, and that this free reconstruction lifts the linear/non-linear structure from values to sheaves. In the wider programme, this fixes the operational side of a conjectured larger adjunction between concurrent practice and modal foundations; the present theorem makes that operational side concrete, with descent, sheafification, and the linear/non-linear structure all machine-checked.

Limitations

We are explicit about what is *not* done here.

- The value-level adjunction $F \dashv G$ is *assumed*, not constructed.
- No concrete session-type model is yet exhibited for V together with a verification that it has the required finite products; the theorem is generic in its hypotheses.
- The topology is the event-discrete one: causal order and overlaps of events do not yet carry descent content. Enriching the site so that they do is the natural next step, and is where colimit and exactness questions return.
- Point sheaves are defined by descent for the explicitly presented single-event covers; we do not yet prove that they coincide with the sheaves for a fully formalized Grothendieck coverage, nor compare against a library coverage.
- This is not the cross-site adjunction to the modal/foundational side; it is the first checked result on the operational side alone.

A note on authorship

This abstract was drafted with the assistance of a language model and reviewed by the authors; the surrounding framework and its philosophical motivation are presented separately. The mathematics it reports, by contrast, is not provisional: every claim above is backed by a proof that the cubical Agda type-checker accepts, without recourse to postulates in the relevant modules.

References

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